Question 2 — Amicable Number Pairs

Sample input: n = 300. The program checks pairs up to 300 and prints a pair when both numbers are amicable. Two numbers are amicable if the sum of the proper divisors of each number equals the other number.

Function: sumdivisors(x) checks integers from 1 through x/2. If x % i == 0, it adds i to sum.

1. Trace of sumdivisors(220)

The function starts with sum = 0. It adds each proper divisor of 220.

Divisor i	220 % i == 0?	sum after adding i
1	Yes	1
2	Yes	3
4	Yes	7
5	Yes	12
10	Yes	22
11	Yes	33
20	Yes	53
22	Yes	75
44	Yes	119
55	Yes	174
110	Yes	284

Returned value: 284.

2. Trace of sumdivisors(284)

The function starts with sum = 0 and checks i from 1 through 142.

Divisor i	284 % i == 0?	sum after adding i
1	Yes	1
2	Yes	3
4	Yes	7
71	Yes	78
142	Yes	220

Returned value: 220.

3. Main program trace for the amicable pair

For a = 220, the program calculates b = sumdivisors(220) = 284. It then checks b > a and b <= n. Both conditions are true because 284 > 220 and 284 <= 300. Next it calculates sumdivisors(284), which returns 220. Since this equals a, the pair is printed and count is increased.

Step	a	b	Condition / operation	count
1	220	—	b = sumdivisors(220) = 284	0
2	220	284	b > a is true; b <= 300 is true	0
3	220	284	sumdivisors(284) == 220 is true	1
4	220	284	Print (220, 284)	1
5	284	220	b > a is false; pair is not printed again	1

Final output for n = 300:
Amicable pair: (220, 284)

Total amicable pairs found: 1.

Why is the pair printed only once? The condition b > a ensures the program prints (220, 284), but does not print the reverse pair (284, 220).

Note: This is a sample dry run using n = 300. If your assignment/test input uses a different limit, the trace and output may differ.